Machine Learning Department
School of Computer Science
Carnegie Mellon University

Diffusion Models

Aran Nayebi & Matt Gormley
Lecture 7

Reminders

- Homework 1: Generative Models of Text
  - Out: Mon, Jan 26
  - Slot A Due (no AI use!): Mon, Feb 9 at 11:59pm
- Quiz 2:
  - In-class: Mon, Feb 16
  - Lectures 5-9: Computer Vision - VAEs
- Homework 2: Generative Models of Images
  - Out: Mon, Feb 9
  - Slot A Due: Sat, Feb 21 at 11:59pm

UNSUPERVISED LEARNING

Unsupervised Learning

Recall…

Assumptions:

1. our data comes from some distribution $p^*(\mathbf{x}_0)$
2. we choose a distribution $p_\theta(\mathbf{x}_0)$ for which sampling $x_0 \sim p_\theta(\mathbf{x}_0)$ is tractable

Goal: learn θ s.t. $p_\theta(\mathbf{x}_0) \approx p^*(\mathbf{x}_0)$

Unsupervised Learning

Recall…

Assumptions:

1. our data comes from some distribution $p^*(\mathbf{x}_0)$
2. we choose a distribution $p_\theta(\mathbf{x}_0)$ for which sampling $x_0 \sim p_\theta(\mathbf{x}_0)$ is tractable

Goal: learn θ s.t. $p_\theta(\mathbf{x}_0) \approx p^*(\mathbf{x}_0)$

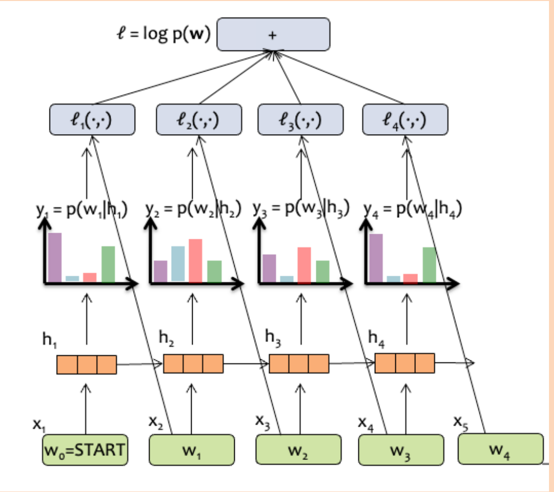
Example: autoregressive LMs

- true $p^*(\mathbf{x}_0)$ is the (human) process that produced text on the web
- choose $p_\theta(\mathbf{x}_0)$ to be an autoregressive language model
  - autoregressive structure means that $p(\mathbf{x}_t \mid \mathbf{x}_1, \ldots, \mathbf{x}_{t-1}) \sim$ Categorical(.) and ancestral sampling is exact/efficient
- learn by finding

  $$\theta \approx \mathrm{argmax}_\theta \log(p_\theta(\mathbf{x}_0))$$

  using gradient based updates on

  $$\nabla_\theta \log(p_\theta(\mathbf{x}_0))$$



Unsupervised Learning

Recall...

Assumptions:

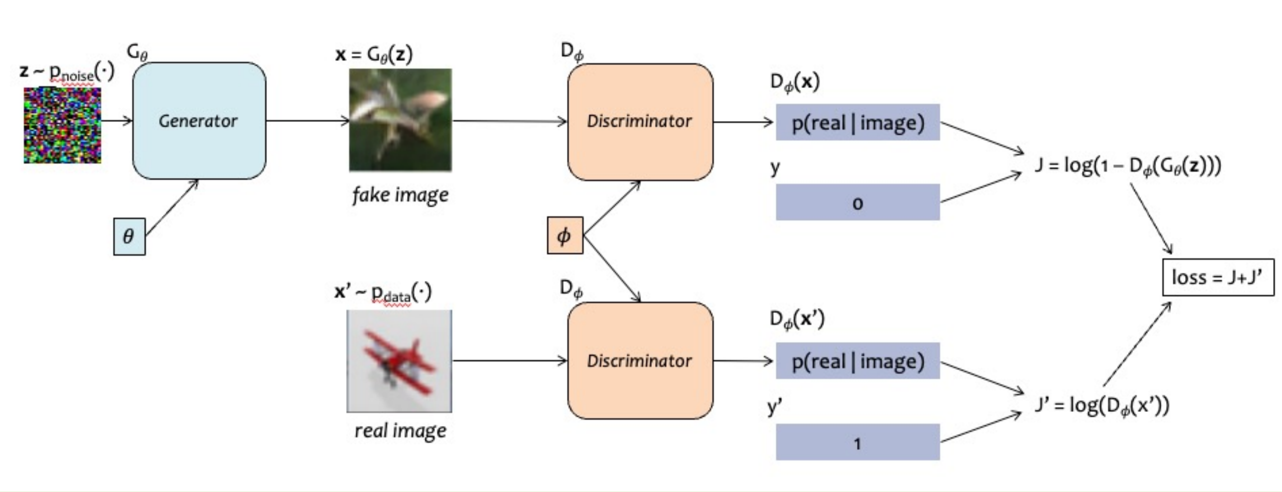
1. our data comes from some distribution $p^*(\mathbf{x}_0)$
2. we choose a distribution $p_\theta(\mathbf{x}_0)$ for which sampling $x_0 \sim p_\theta(\mathbf{x}_0)$ is tractable

Goal: learn θ s.t. $p_\theta(\mathbf{x}_0) \approx p^*(\mathbf{x}_0)$

Example: GANs

- true $p^*(\mathbf{x}_0)$ is distribution over photos taken and posted to Flikr
- choose $p_\theta(\mathbf{x}_0)$ to be an expressive model (e.g. noise fed into inverted CNN) that can generate images
  - sampling is typically easy: $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ and $\mathbf{x}_0 = f_\theta(\mathbf{z})$
- learn by finding $\theta \approx \text{argmax}_\theta \log(p_\theta(\mathbf{x}_0))$?
  - No! Because we can't even compute $\log(p_\theta(\mathbf{x}_0))$ or its gradient
  - Why not? Even for 1-layer neural networks: *single* image x_0 mapping from noise *z* in conditional makes marginal $p(x_0)$ almost 0 everywhere & difficult to differentiate

$$p_\theta(\mathbf{x}_0) = \int_z p_\theta(\mathbf{x}_0 \mid z)\, p(z)\, dz$$



so optimize a minimax loss instead

Unsupervised Learning

Recall…

Assumptions:

1. our data comes from some distribution $p^*(\mathbf{x}_0)$
2. we choose a distribution $p_\theta(\mathbf{x}_0)$ for which sampling $x_0 \sim p_\theta(\mathbf{x}_0)$ is tractable

Goal: learn θ s.t. $p_\theta(\mathbf{x}_0) \approx p^*(\mathbf{x}_0)$

Example: VAEs / Diffusion Models

- true $p^*(\mathbf{x}_0)$ is distribution over photos taken and posted to Flikr
- choose $p_\theta(\mathbf{x}_0)$ to be an expressive model (e.g. noise fed into inverted CNN) that can generate images
  - sampling is will be easy
- learn by finding $\theta \approx \text{argmax}_\theta \log(p_\theta(\mathbf{x}_0))$?
  - Sort of! We can't compute the gradient $\nabla_\theta \log(p_\theta(\mathbf{x}_0))$
  - So we instead optimize a variational lower bound (more on that later)

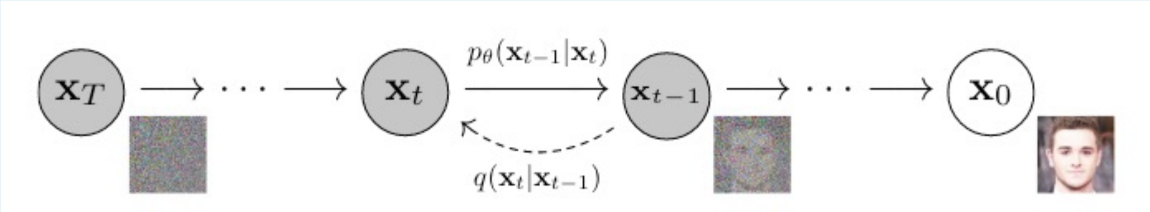


Figure from Ho et al. (2020)

Latent Variable Models

Recall...

- For GANs and VAEs, we assume that there are (unknown) **latent variables** which give rise to our observations
- The **vector z** are those latent variables
- After learning a GAN or VAE, we can **interpolate** between images in latent **z** space

Figure 4: Top rows: Interpolation between a series of 9 random points in Z show that the space learned has smooth transitions, with every image in the space plausibly looking like a bedroom. In the 6th row, you see a room without a window slowly transforming into a room with a giant window. In the 10th row, you see what appears to be a TV slowly being transformed into a window.

Figure from Radford et al. (2016)

GAN → VAE → Diffusion

GAN → VAE → Diffusion

GANs

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$

$p_\theta(\mathbf{x} \mid \mathbf{z})$ = point mass on $G_\theta(\mathbf{z})$

Model:

$$p_\theta(\mathbf{x} \mid \mathbf{z}) = \begin{cases} 1 & \text{if } \mathbf{x} = G_\theta(\mathbf{z}) \\ 0 & \text{otherwise} \end{cases}$$

MLE?:

$$\hat{\theta} = \arg\max_\theta \sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)})$$

NOPE!

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})\, d\mathbf{z}$$

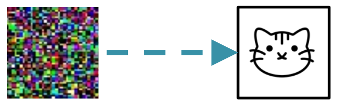
Learn: minimax game

GAN → VAE → Diffusion

GANs

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$ $\quad p_\theta(\mathbf{x} \mid \mathbf{z}) =$ point mass on $G_\theta(\mathbf{z})$

Model:



$$p_\theta(\mathbf{x} \mid \mathbf{z}) = \begin{cases} 1 & \text{if } \mathbf{x} = G_\theta(\mathbf{z}) \\ 0 & \text{otherwise} \end{cases}$$

MLE?: $\hat{\theta} = \arg\max_\theta \sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)})$ NOPE!

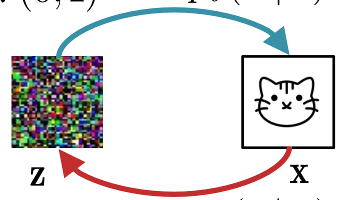
$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})\, d\mathbf{z}$$

Learn: minimax game

VAEs

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$ $\quad p_\theta(\mathbf{x} \mid \mathbf{z}) \sim \mathcal{N}(\mu_\theta(\mathbf{z}), \Sigma_\theta(\mathbf{z}))$

Model:



MLP

$q_\phi(\mathbf{z} \mid \mathbf{x}) \sim \mathcal{N}(\mu_\phi(\mathbf{x}), \Sigma_\phi(\mathbf{x}))$

$h = \tanh(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1)$

$\mu_\theta(\mathbf{z}) = \mathbf{W}_2 h + \mathbf{b}_2$

$\Sigma_\theta(\mathbf{z}) = (\mathbf{W}_3 h + \mathbf{b}_3)\mathbf{I}$

MLE?: $\hat{\theta} = \arg\max_\theta \sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)})$ NOPE!

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})\, d\mathbf{z}$$

Learn: $\theta, \phi = \arg\min_{\theta,\phi} \mathrm{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p_\theta(\mathbf{z} \mid \mathbf{x}))$

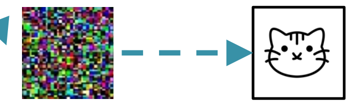
where $p_\theta(\mathbf{z} \mid \mathbf{x})$ is true posterior of $p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})$

GAN → VAE → Diffusion

GANs

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$ $\quad p_\theta(\mathbf{x} \mid \mathbf{z}) =$ point mass on $G_\theta(\mathbf{z})$

Model:



$\mathbf{z}$ $\quad \mathbf{x}$

$$p_\theta(\mathbf{x} \mid \mathbf{z}) = \begin{cases} 1 & \text{if } \mathbf{x} = G_\theta(\mathbf{z}) \\ 0 & \text{otherwise} \end{cases}$$

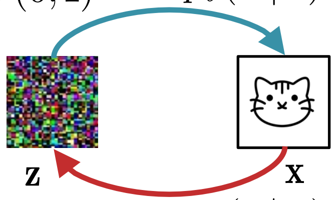
MLE?:
$$\hat{\theta} = \arg\max_\theta \sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)})$$
NOPE!
$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})\, d\mathbf{z}$$

Learn: minimax game

VAEs

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$ $\quad p_\theta(\mathbf{x} \mid \mathbf{z}) \sim \mathcal{N}(\mu_\theta(\mathbf{z}), \Sigma_\theta(\mathbf{z}))$

Model:



$\mathbf{z}$ $\quad \mathbf{x}$

MLP

$q_\phi(\mathbf{z} \mid \mathbf{x}) \sim \mathcal{N}(\mu_\phi(\mathbf{x}), \Sigma_\phi(\mathbf{x}))$

$$h = \tanh(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1)$$
$$\mu_\theta(\mathbf{z}) = \mathbf{W}_2 h + \mathbf{b}_2$$
$$\Sigma_\theta(\mathbf{z}) = (\mathbf{W}_3 h + \mathbf{b}_3)\mathbf{I}$$

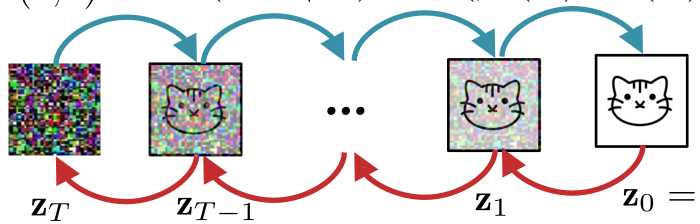
MLE?:
$$\hat{\theta} = \arg\max_\theta \sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)})$$
NOPE!
$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})\, d\mathbf{z}$$

Learn:
$$\theta, \phi = \arg\min_{\theta,\phi} \mathrm{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p_\theta(\mathbf{z} \mid \mathbf{x}))$$
where $p_\theta(\mathbf{z} \mid \mathbf{x})$ is true posterior of $p_\theta(\mathbf{x} \mid \mathbf{z})\, p_\theta(\mathbf{z})$

Diffusion

$p_\theta(\mathbf{z}) \sim \mathcal{N}(0, \mathbf{I})$ $\quad p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{z}_t), \Sigma_\theta(\mathbf{z}_t))$

Model:



$\mathbf{z}_T$ $\quad \mathbf{z}_{T-1}$ $\quad \mathbf{z}_1$ $\quad \mathbf{z}_0 = \mathbf{x}$

$q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1}) \sim \mathcal{N}(\mu_\phi(\mathbf{z}_{t-1}), \Sigma_\phi(\mathbf{z}_{t-1}))$ $\quad q_\phi(\mathbf{z}_0) =$ data dist.

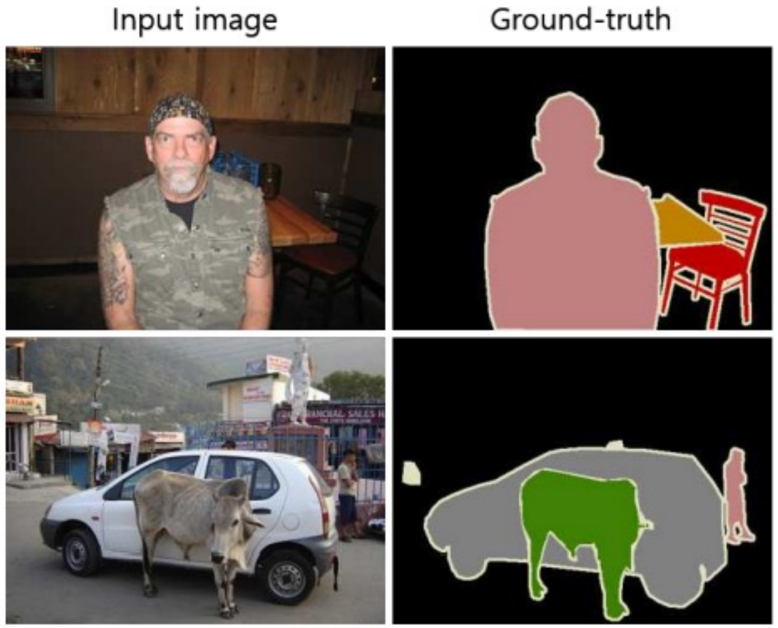
Learn:
$$\theta = \arg\min_\theta \mathrm{KL}(q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}) \,\|\, p_\theta(\mathbf{z}_{1:T} \mid \mathbf{x})), \qquad \phi \text{ chosen ahead of time}$$
where $p_\theta(\mathbf{z}_{1:T} \mid \mathbf{x})$ is true posterior of $p_\theta(\mathbf{x} \mid \mathbf{z}_1) \cdots p_\theta(\mathbf{z}_{T-1} \mid \mathbf{z}_T)\, p_\theta(\mathbf{z}_T)$

MLE?: NOPE!

U-NET

Semantic Segmentation

- Given an image, predict a label for every pixel in the image
- Not merely a classification problem, because there are strong correlations between pixel-specific labels



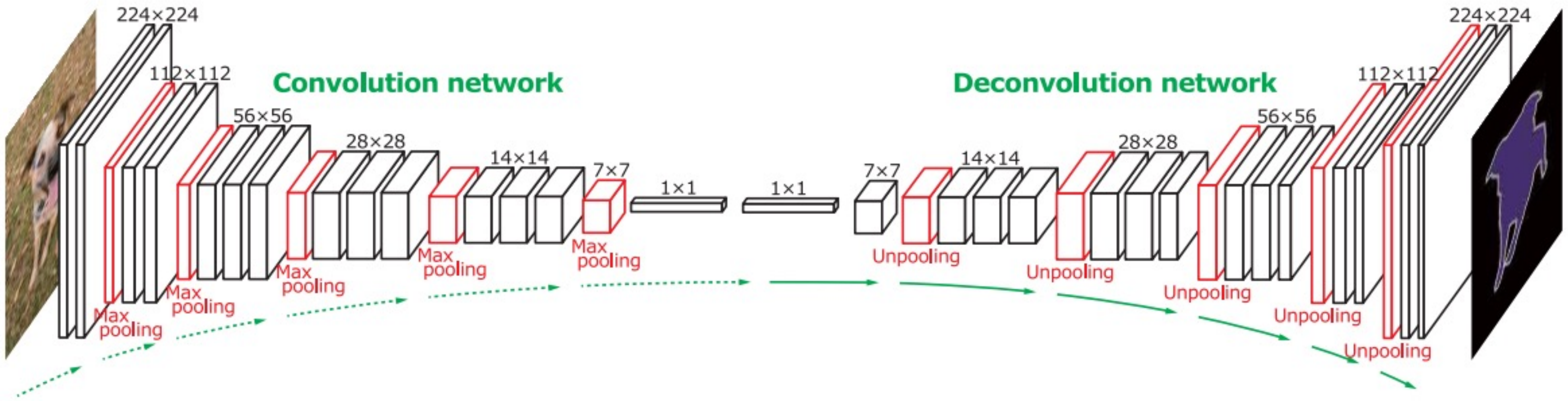


Figure from https://openaccess.thecvf.com/content_iccv_2015/papers/Noh_Learning_Deconvolution_Network_ICCV_2015_paper.pdf

Instance Segmentation

- Predict per-pixel labels as in semantic segmentation, but differentiate between different instances of the same label
- *Example*: if there are two people in the image, one person should be labeled **person-1** and one should be labeled **person-2**

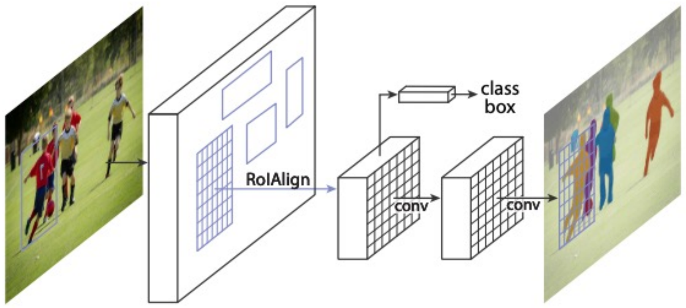
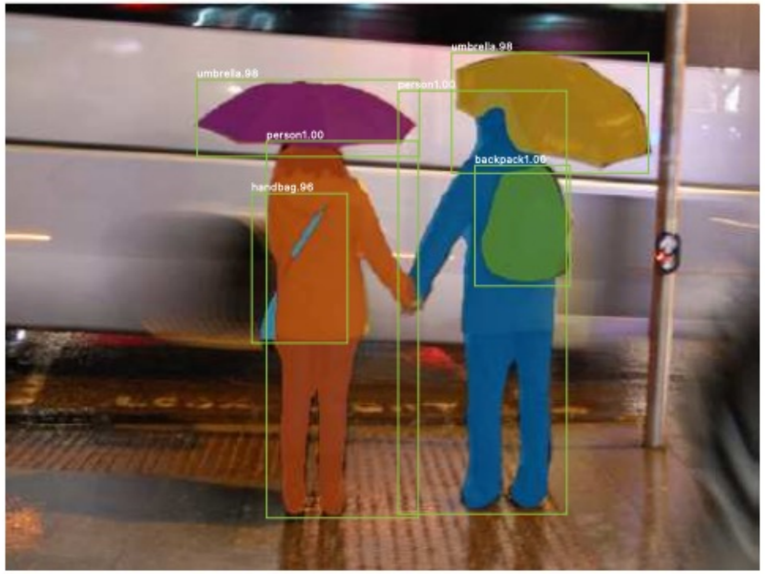


Figure 1. The **Mask R-CNN** framework for instance segmentation.



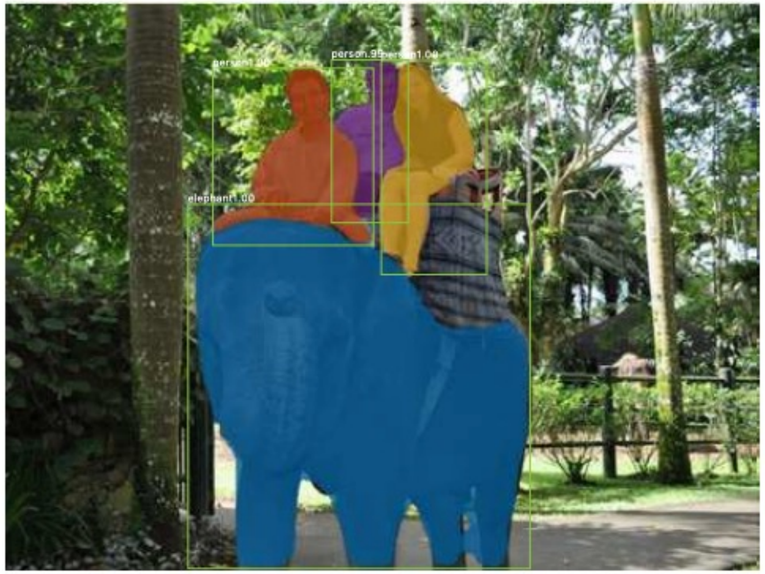


Figure from https://openaccess.thecvf.com/content_ICCV_2017/papers/He_Mask_R-CNN_ICCV_2017_paper.pdf

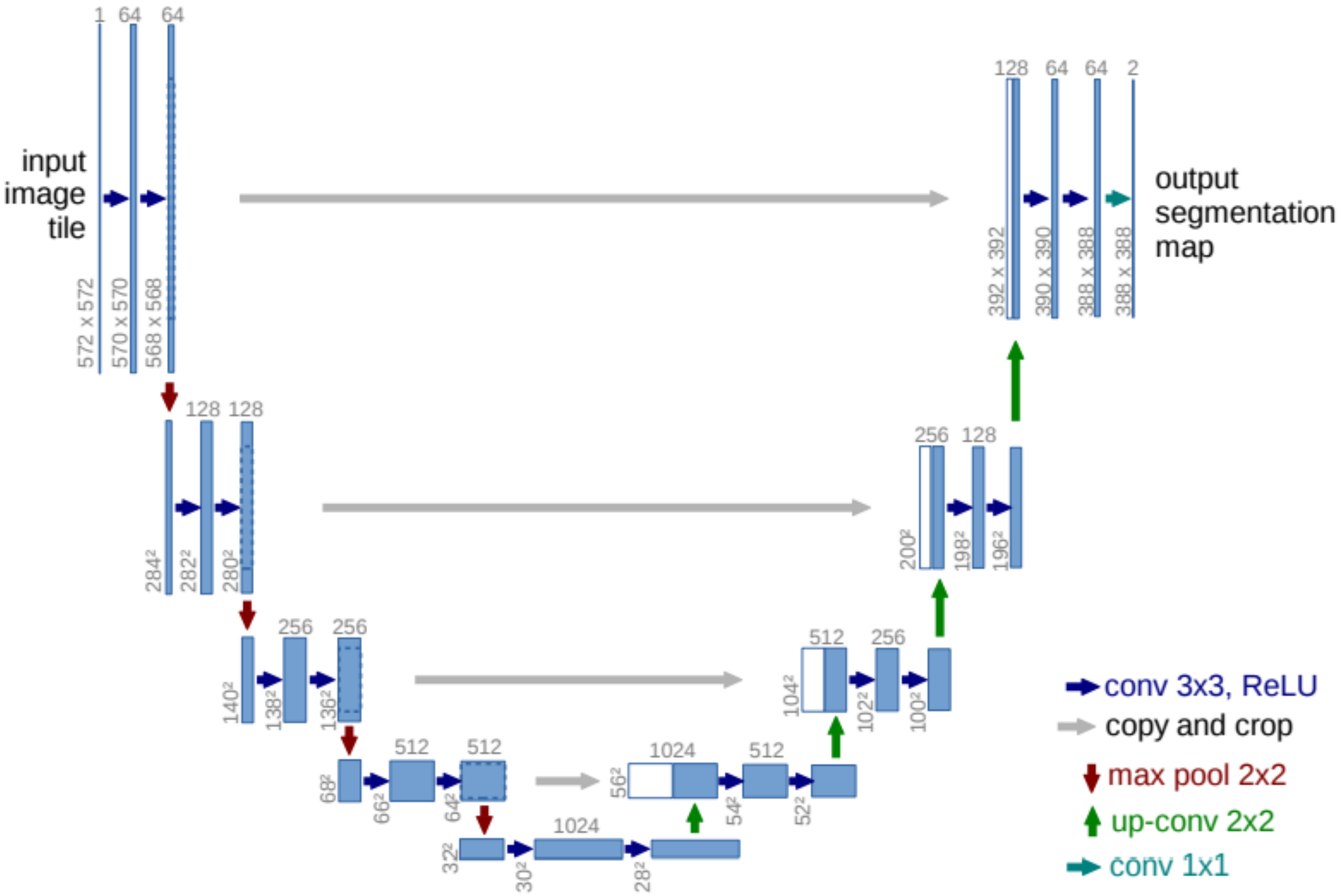
U-Net

Contracting path

- block consists of:
  - 3x3 convolution
  - 3x3 convolution
  - ReLU
  - max-pooling with stride of 2 (downsample)
- repeat the block N times, doubling number of channels

Expanding path

- block consists of:
  - 2x2 convolution (upsampling)
  - concatenation with contracting path features
  - 3x3 convolution
  - 3x3 convolution
  - ReLU
- repeat the block N times, halving the number of channels



U-Net

- Originally designed for applications to biomedical segmentation
- Key observation is that the output layer has the **same** dimensions as the input image (possibly with different number of channels)

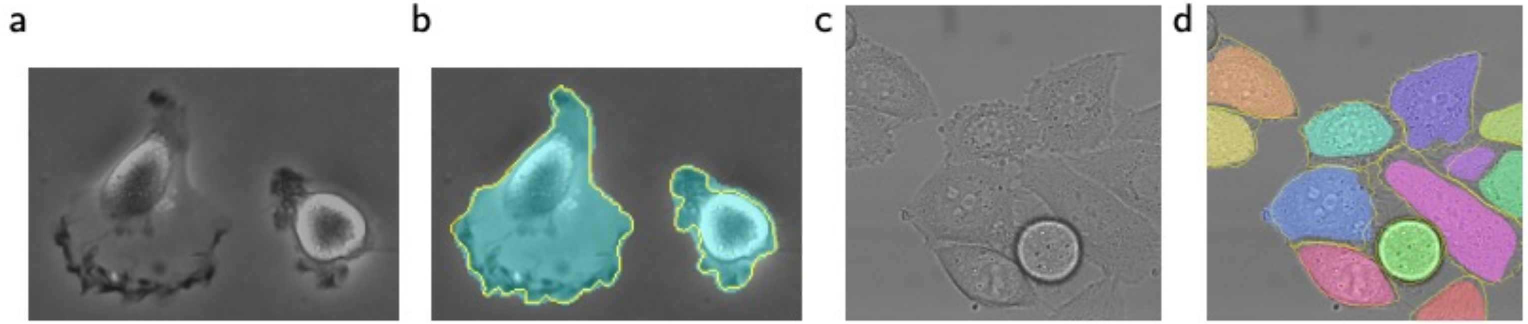
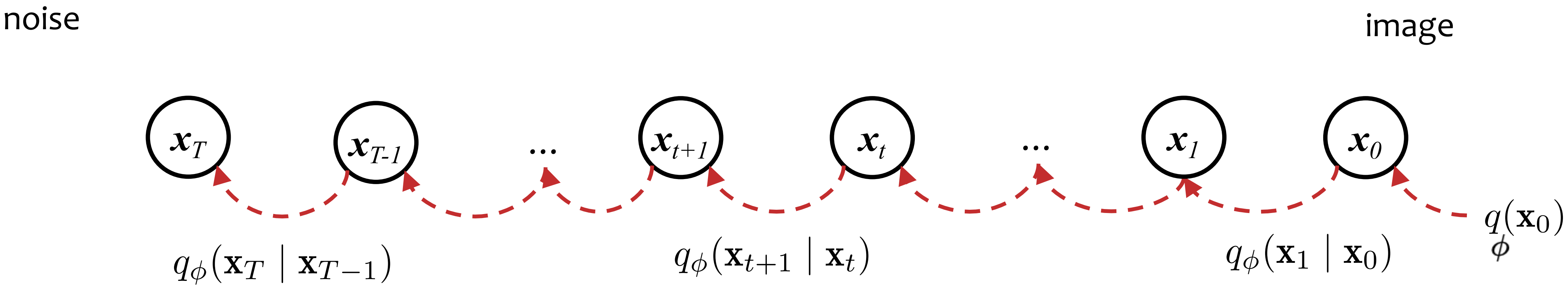


Fig. 4. Result on the ISBI cell tracking challenge. (**a**) part of an input image of the "PhC-U373" data set. (**b**) Segmentation result (cyan mask) with manual ground truth (yellow border) (**c**) input image of the "DIC-HeLa" data set. (**d**) Segmentation result (random colored masks) with manual ground truth (yellow border).

DIFFUSION MODELS

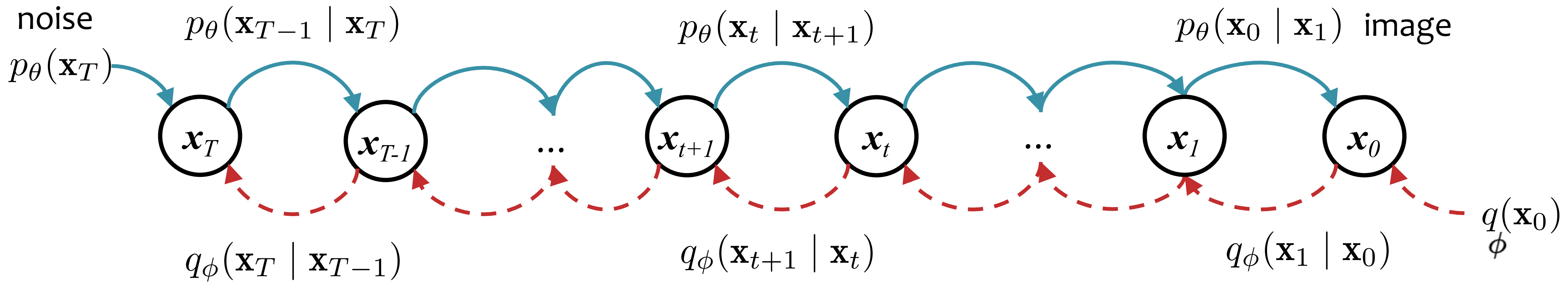
Diffusion Model



Forward Process:

$$q_\phi(\mathbf{x}_{0:T}) = \underset{\phi}{q}(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

Diffusion Model



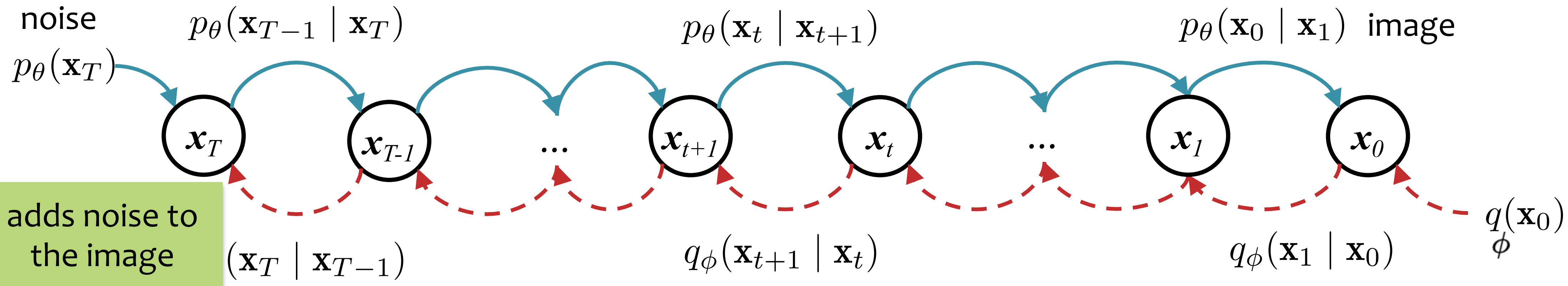
Forward Process:

$$q_\phi(\mathbf{x}_{0:T}) = \underset{\phi}{q}(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

(Learned) Reverse Process:

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

Diffusion Model



Forward Process:

if we could sample from this we'd be done

$$q_\phi(\mathbf{x}_{0:T}) = \boxed{q(\mathbf{x}_0)}_{\phi} \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

(Learned) Reverse Process:

removes noise

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} \boxed{p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)}$$

goal is to learn this

Diffusion Model

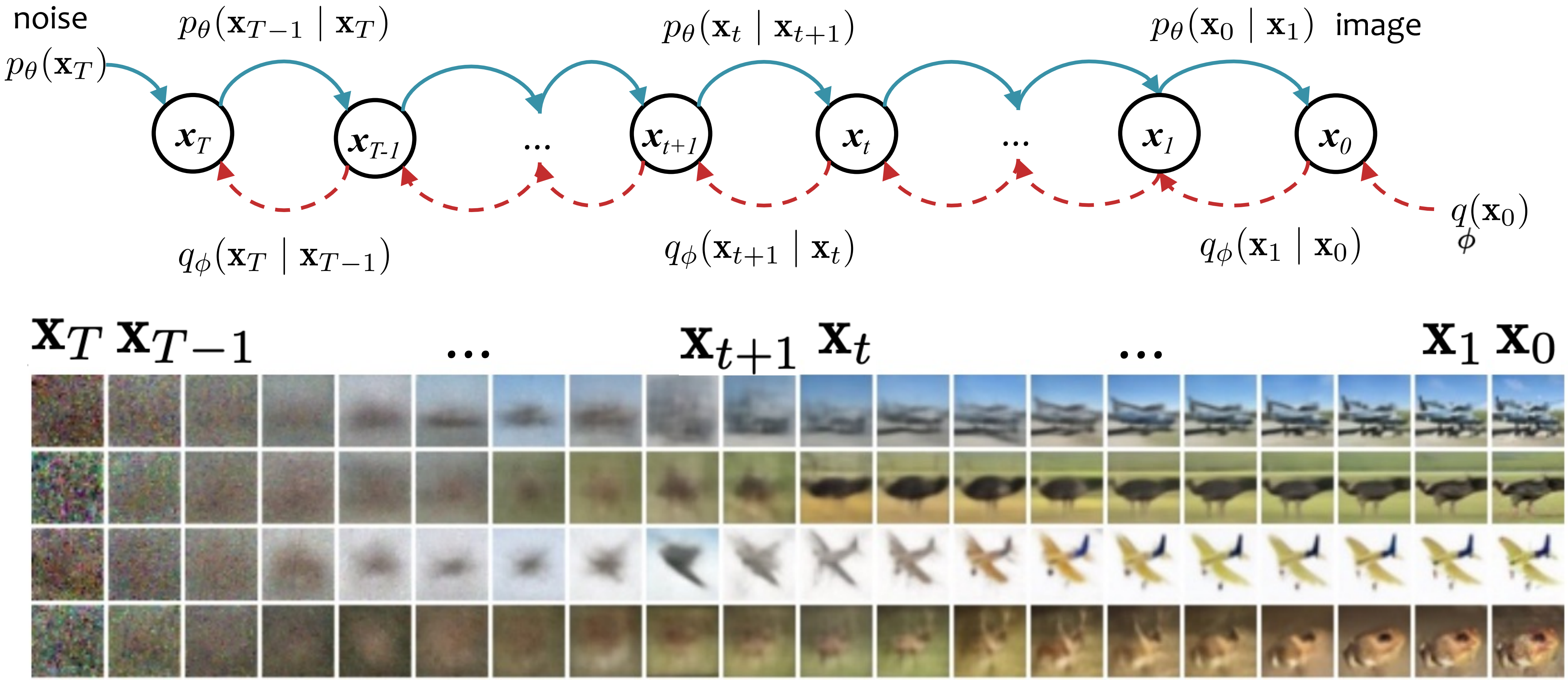
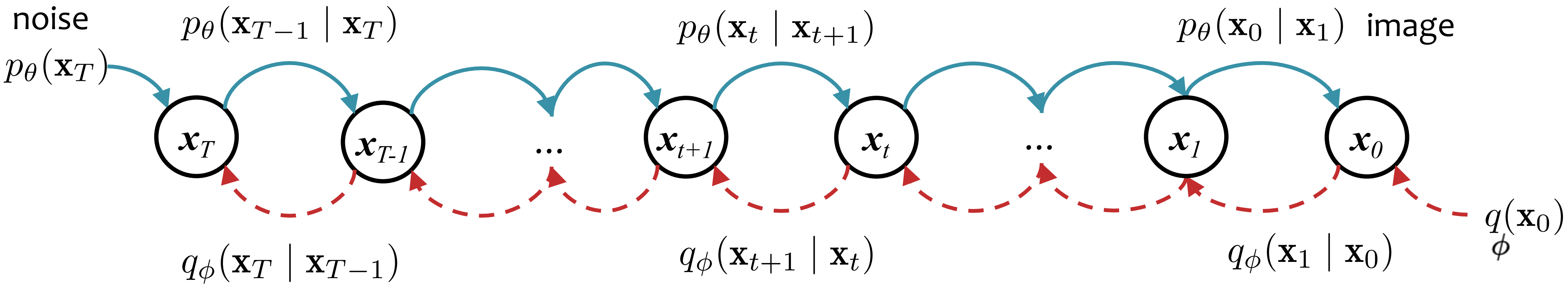
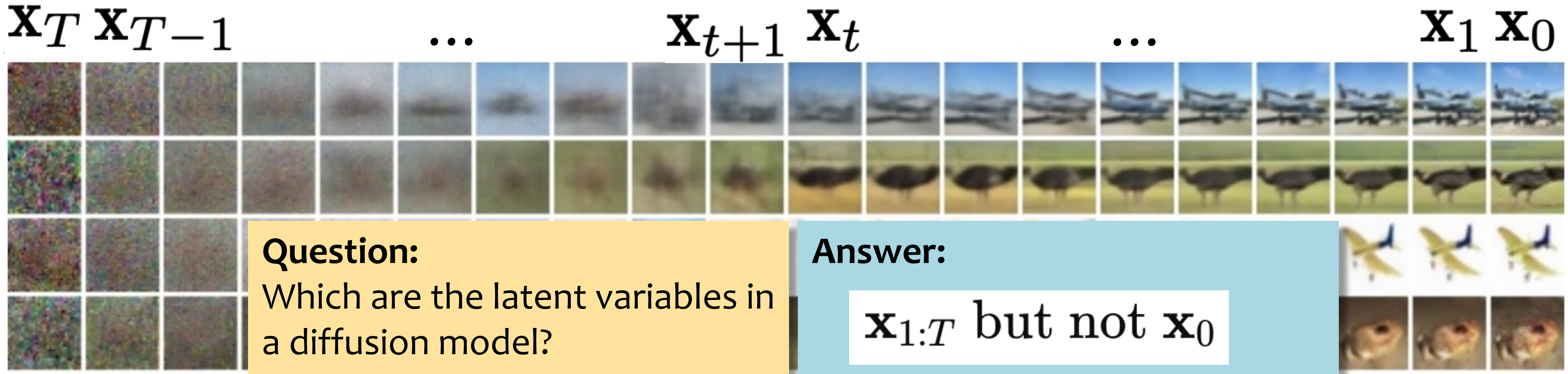


Figure from Ho et al. (2020)

Diffusion Model





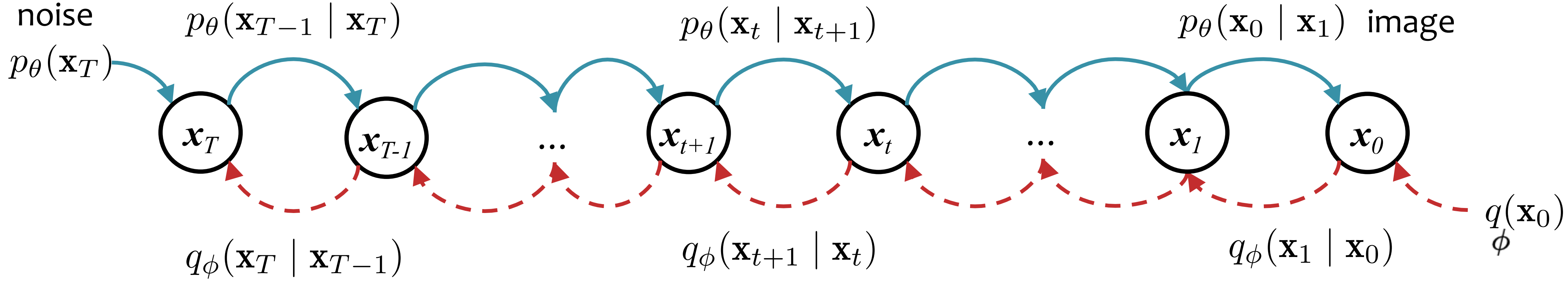
Question:
Which are the latent variables in a diffusion model?

Answer:

$\mathbf{x}_{1:T}$ but not $\mathbf{x}_0$

Figure from Ho et al. (2020)

Denoising Diffusion Probabilistic Model (DDPM)



Forward Process (*adds* noise to image):

$$q_\phi(\mathbf{x}_{0:T}) = q(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

$q(\mathbf{x}_0)$ = data distribution

$$q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

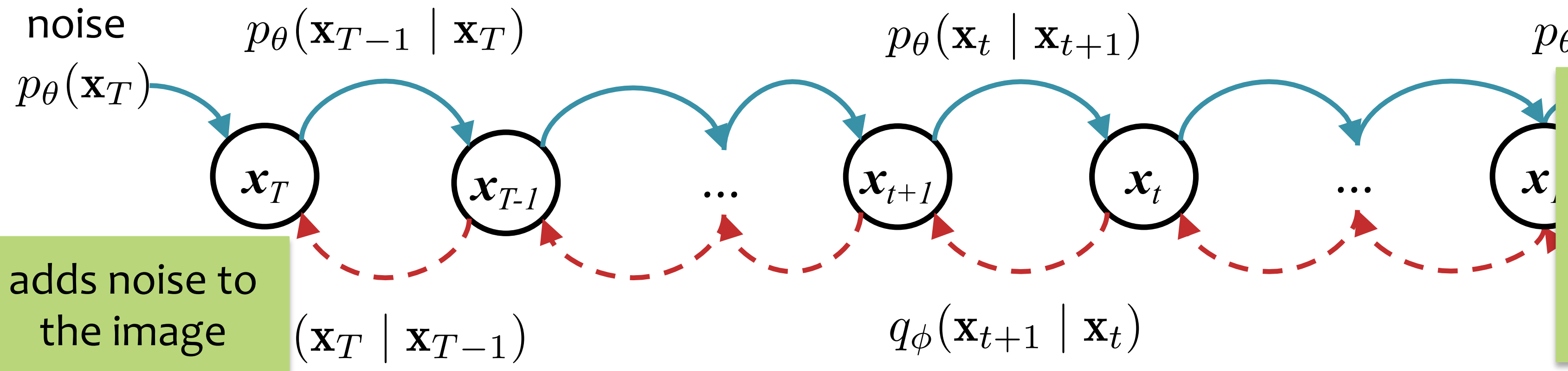
(Learned) Reverse Process (*removes* noise):

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

$$p_\theta(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$$

Diffusion Model



learning is hard.
why don't we instead just infer the *exact* reverse process corresponding to the forward process?

adds noise to the image

Forward Process:

if we could sample from this we'd be done

$$q_\phi(\mathbf{x}_{0:T}) = \boxed{q_\phi(\mathbf{x}_0)} \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

(Learned) Reverse Process:

removes noise

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} \boxed{p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)}$$

goal is to learn this

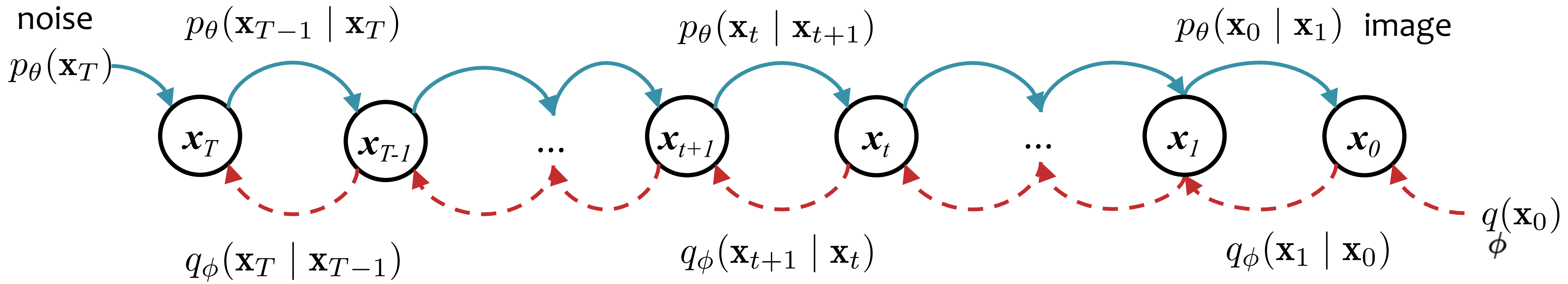
(Exact) Reverse Process:

$$\boxed{q_\phi(\mathbf{x}_{0:T}) = q_\phi(\mathbf{x}_T)} \prod_{t=1}^{T} q_\phi(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

The *exact* reverse process requires inference. And, even though $q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ is simple, computing $q_\phi(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is intractable! Why? Because $q(\mathbf{x}_0)$ might be not-so-simple.

$$q_\phi(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \frac{q_\phi(\mathbf{x}_{t-1}, \mathbf{x}_t)}{q_\phi(\mathbf{x}_t)} = \frac{\int q_\phi(\mathbf{x}_{0:T})\, d\mathbf{x}_{0:t-2}\, d\mathbf{x}_{t+1:T}}{\int q_\phi(\mathbf{x}_{0:T})\, d\mathbf{x}_{0:t-1}\, d\mathbf{x}_{t+1:T}}$$

Denoising Diffusion Probabilistic Model (DDPM)



Forward Process:

$$q_\phi(\mathbf{x}_{0:T}) = \underset{\phi}{q}(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

$$\underset{\phi}{q}(\mathbf{x}_0) = \text{data distribution}$$

$$q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

(Learned) Reverse Process:

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

$$p_\theta(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$$

Defining the Forward Process

Forward Process:

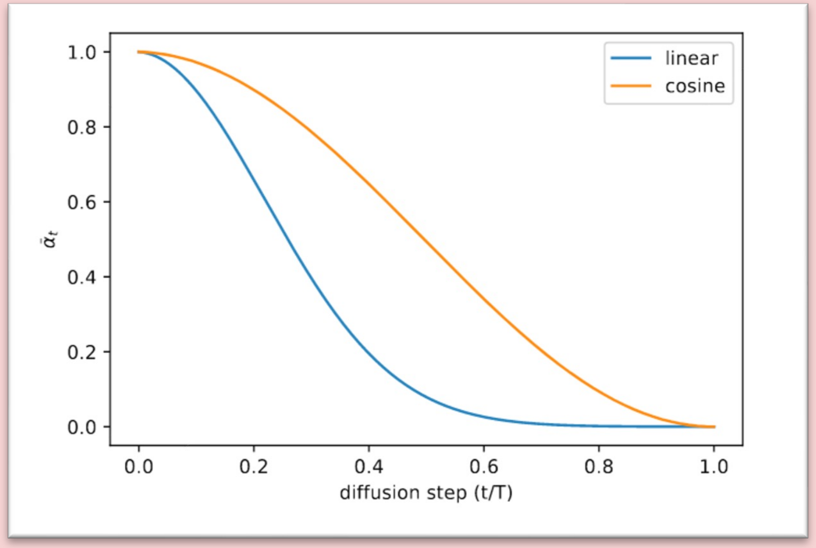
$$q_\phi(\mathbf{x}_{0:T}) = q(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

$q(\mathbf{x}_0) =$ data distribution

$$q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

Noise schedule:

We choose α_t to follow a fixed schedule s.t. $q_\phi(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, just like $p_\theta(\mathbf{x}_T)$.



Gaussians (an aside)

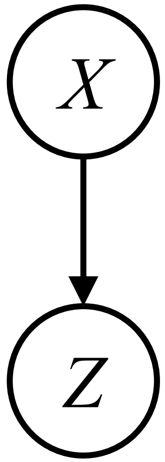
Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$

1. Sum of two Gaussians is a Gaussian. $X+Y \sim \mathcal{N}(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$
2. Difference of two Gaussians is a Gaussian. $X-Y \sim \mathcal{N}(\mu_x-\mu_y, \sigma_x^2+\sigma_y^2)$

Let $Z \mid X \sim \mathcal{N}(\mu_z = g(X), \sigma_z^2)$ *(i.e.,* $Z = g(X) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma_z^2)$*)*

3. Gaussian with a Gaussian mean passed through a *linear/affine* function $g(\cdot)$ has a Gaussian marginal,

$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$ is a Gaussian density



Defining the Forward Process

Forward Process:

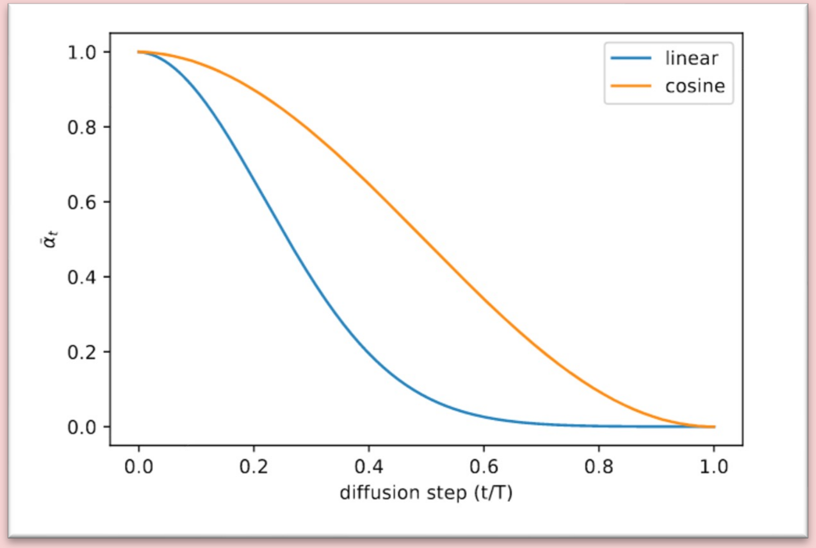
$$q_\phi(\mathbf{x}_{0:T}) = q(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

$$q(\mathbf{x}_0) = \text{data distribution}$$

$$q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

Noise schedule:

We choose α_t to follow a fixed schedule s.t. $q_\phi(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, just like $p_\theta(\mathbf{x}_T)$.



Property #1:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

$$\text{where } \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

Q: So what is $q_\phi(\mathbf{x}_T \mid \mathbf{x}_0)$? Note the *capital* T in the subscript.

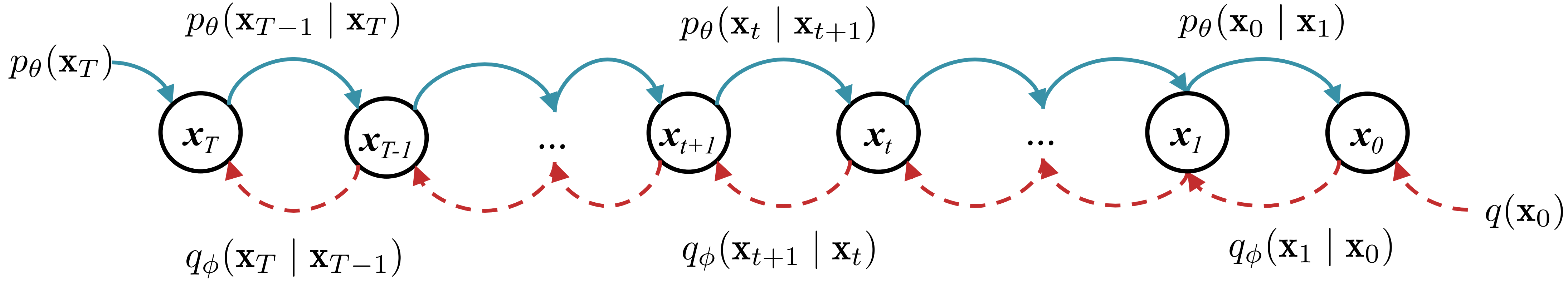
A:

$$\sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathbf{x}_t = \sqrt{\alpha_t}\,\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Diffusion Model



Forward Process:

$$q_\phi(\mathbf{x}_{0:T}) = \underset{\phi}{q}(\mathbf{x}_0) \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

(Learned) Reverse Process:

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

Q: If q_ϕ is just adding noise, how can p_θ be interesting at all?

A:

Q: But if $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is Gaussian, how can it learn a θ such that $p_\theta(\mathbf{x}_0) \approx q(\mathbf{x}_0)$? Won't $p_\theta(\mathbf{x}_0)$ be Gaussian too?

A:

Gaussians (an aside)

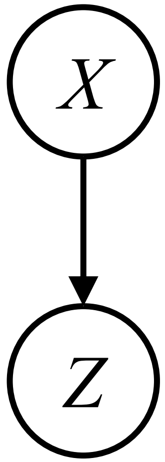
Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$

1. Sum of two Gaussians is a Gaussian. $X+Y \sim \mathcal{N}(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$
2. Difference of two Gaussians is a Gaussian. $X-Y \sim \mathcal{N}(\mu_x-\mu_y, \sigma_x^2+\sigma_y^2)$

Let $Z \mid X \sim \mathcal{N}(\mu_z = g(X), \sigma_z^2)$ *(i.e.,* $Z = g(X) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma_z^2)$*)*

3. Gaussian with a Gaussian mean passed through a *linear/affine* function $g(\cdot)$ has a Gaussian marginal,

$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$ is a Gaussian density



Gaussians (an aside)

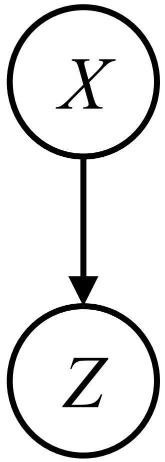
Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$

1. Sum of two Gaussians is a Gaussian. $X+Y \sim \mathcal{N}(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$
2. Difference of two Gaussians is a Gaussian. $X-Y \sim \mathcal{N}(\mu_x-\mu_y, \sigma_x^2+\sigma_y^2)$

Let $Z \mid X \sim \mathcal{N}(\mu_z = g(X), \sigma_z^2)$ *(i.e., $Z = g(X) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma_z^2)$)*

3. Gaussian with a Gaussian mean passed through a *linear/affine* function $g(\cdot)$ has a Gaussian marginal,

$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$ is a Gaussian density



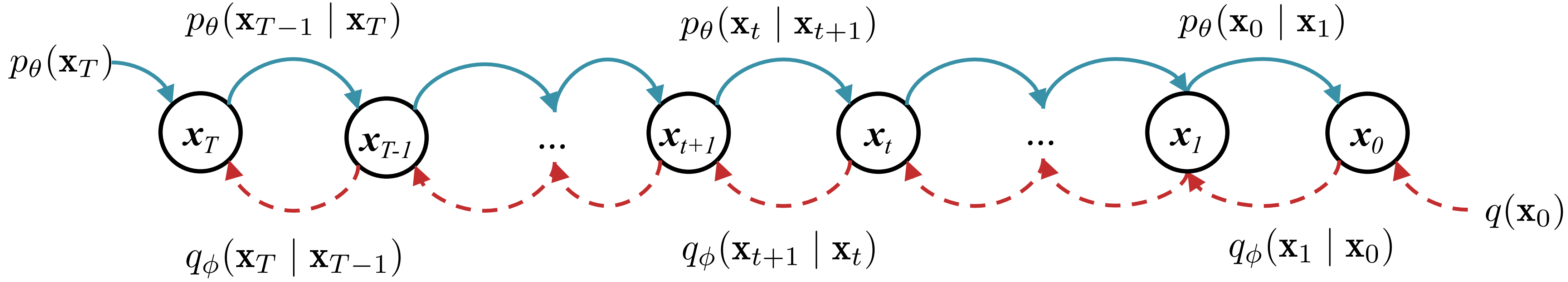
4. Gaussian **conditional** with a Gaussian parent passed through a nonlinear function $g(\cdot)$ does not have a Gaussian marginal **after marginalizing out** X

$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$ is **not** a Gaussian density in general

$$\mathbf{x}_t = \sqrt{\alpha_t}\,\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Diffusion Model



Forward Process:

$$q_\phi(\mathbf{x}_{0:T}) = \underset{\phi}{q(\mathbf{x}_0)} \prod_{t=1}^{T} q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

(Learned) Reverse Process:

$$p_\theta(\mathbf{x}_{0:T}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

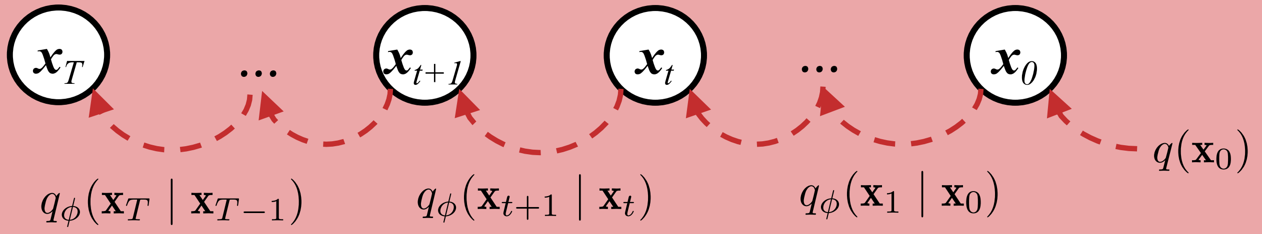
Q: If q_ϕ is just adding noise, how can p_θ be interesting at all?

A: Because $q(\mathbf{x}_0)$ is not just a noise distribution and p_θ must capture that interesting variability

Q: But if $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is Gaussian, how can it learn a θ such that $p_\theta(\mathbf{x}_0) \approx q(\mathbf{x}_0)$? Won't $p_\theta(\mathbf{x}_0)$ be Gaussian too?

A:

Asymmetry of Forward/Reverse Processes



In the forward process, if we marginalize out the intermediate variables of the Markov chain, we get back a Gaussian:

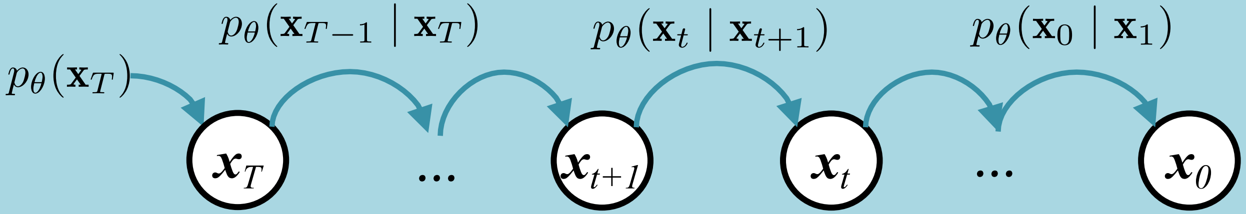
$$\underset{\phi}{q}(\mathbf{x}_t \mid \mathbf{x}_0) = \int \left[\prod_{s=1}^{t} q(\mathbf{x}_s \mid \mathbf{x}_{s-1}) \right] d\mathbf{x}_{1:t-1}$$
$$= \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

Forward Process:

$$\underset{\phi}{q}(\mathbf{x}_0) = \text{data distribution}$$
$$q_\phi(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

(Learned) Reverse Process:

$$p_\theta(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$
$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$$

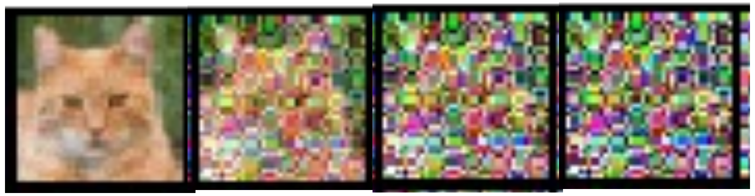


In the reverse process, by contrast, if we marginalize out the intermediate variables of the Markov chain, we do NOT get a Gaussian.

$$p_\theta(\mathbf{x}_0) = \int \left[p_\theta(\mathbf{x}_T) \prod_{s=1}^{T} p_\theta(\mathbf{x}_{s-1} \mid \mathbf{x}_s) \right] d\mathbf{x}_{1:T}$$

$\neq$ a Gaussian density

Diffusion Model Analogy



Properties of forward and *exact* reverse processes

Property #1:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$

$\Rightarrow$ we can sample $\mathbf{x}_t$ from $\mathbf{x}_0$ at any timestep t efficiently in closed form

$\Rightarrow$ $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

this is the same reparameterization trick from VAEs

Gaussians (an aside)

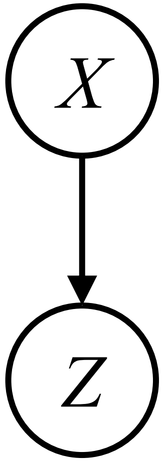
Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$

1. Sum of two Gaussians is a Gaussian. $X+Y \sim \mathcal{N}(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$
2. Difference of two Gaussians is a Gaussian. $X-Y \sim \mathcal{N}(\mu_x-\mu_y, \sigma_x^2+\sigma_y^2)$

Let $Z \mid X \sim \mathcal{N}(\mu_z = g(X), \sigma_z^2)$ *(i.e.,* $Z = g(X) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma_z^2)$*)*

3. Gaussian with a Gaussian mean passed through a *linear/affine* function $g(\cdot)$ has a Gaussian marginal,

$$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$$ is a Gaussian density



4. Gaussian **conditional** with a Gaussian parent passed through a nonlinear function $g(\cdot)$ does not have a Gaussian marginal **after marginalizing out** X

$$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx$$ is **not** a Gaussian density in general

Gaussians (an aside)

Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$

1. Sum of two Gaussians is a Gaussian. $X + Y \sim \mathcal{N}(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$
2. Difference of two Gaussians is a Gaussian. $X - Y \sim \mathcal{N}(\mu_x - \mu_y, \sigma_x^2 + \sigma_y^2)$

Let $Z \mid X \sim \mathcal{N}(\mu_z = g(X), \sigma_z^2)$ *(i.e., $Z = g(X) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma_z^2)$)*

3. Gaussian with a Gaussian mean passed through a *linear/affine* function $g(\cdot)$ has a Gaussian marginal,

   $$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx \text{ is a Gaussian density}$$

   and a Gaussian posterior.

   $$p_{X|Z}(x \mid z) = p_{Z|X}(z|x) p_X(x) / p_Z(z) \text{ is a Gaussian density}$$

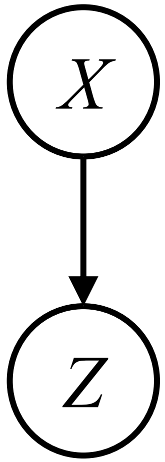
   (Gaussianity holds only when g is linear / affine)

4. Gaussian **conditional** with a Gaussian parent passed through a nonlinear function $g(\cdot)$ does not have a Gaussian marginal **after marginalizing out** X

   $$p_Z(z) = \int_x p_{Z|X}(z|x) p_X(x)\, dx \text{ is \textbf{not} a Gaussian density in general}$$

   nor a Gaussian posterior.

   $$p_{X|Z}(x \mid z) = p_{Z|X}(z|x) p_X(x) / p_Z(z) \text{ is \textbf{not} a Gaussian density in general}$$



Properties of forward and *exact* reverse processes

Property #1:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

$$\text{where } \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$\Rightarrow$ we can sample $\mathbf{x}_t$ from $\mathbf{x}_0$ at any timestep t efficiently in closed form

$\Rightarrow$ $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Property #2: Estimating $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is intractable because of its dependence on $q(\mathbf{x}_0)$. However, conditioning on $\mathbf{x}_0$ we can efficiently work with:

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2 \mathbf{I})$$

$$\text{where } \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$$

$$= \alpha_t^{(0)}\mathbf{x}_0 + \alpha_t^{(t)}\mathbf{x}_t$$

$$\sigma_t^2 = \frac{(1-\bar{\alpha}_{t-1})(1-\alpha_t)}{1-\bar{\alpha}_t}$$

Parameterizing the *learned* reverse process

Recall: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$

Later we will show that given a training sample $\mathbf{x}_0$, we want

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

to be as close as possible to

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

Intuitively, this makes sense: if the *learned* reverse process is supposed to subtract away the noise, then whenever we're working with a specific $\mathbf{x}_0$ it should subtract it away exactly as *exact* reverse process would have.

Parameterizing the *learned* reverse process

Recall: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$

Later we will show that given a training sample $\mathbf{x}_0$, we want

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

to be as close as possible to

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

Intuitively, this makes sense: if the *learned* reverse process is supposed to subtract away the noise, then whenever we're working with a specific $\mathbf{x}_0$ it should subtract it away exactly as *exact* reverse process would have.

Idea #1: Rather than learn $\Sigma_\theta(\mathbf{x}_t, t)$ just use what we know about $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \sim \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2 \mathbf{I})$:

$$\Sigma_\theta(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$$

Idea #2: Choose μ_θ based on $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$, i.e. we want $\mu_\theta(\mathbf{x}_t, t)$ to be close to $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$. Here are three ways we could parameterize this:

Option A: Learn a network that approximates $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$ directly from $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$$

where t is treated as an extra feature in UNet

Parameterizing the *learned* reverse process

Recall: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$

Later we will show that given a training sample $\mathbf{x}_0$, we want

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

to be as close as possible to

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

Intuitively, this makes sense: if the *learned* reverse process is supposed to subtract away the noise, then whenever we're working with a specific $\mathbf{x}_0$ it should subtract it away exactly as *exact* reverse process would have.

Idea #1: Rather than learn $\Sigma_\theta(\mathbf{x}_t, t)$ just use what we know about $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \sim \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2 \mathbf{I})$:

$$\Sigma_\theta(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$$

Idea #2: Choose μ_θ based on $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$, i.e. we want $\mu_\theta(\mathbf{x}_t, t)$ to be close to $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$. Here are three ways we could parameterize this:

Option B: Learn a network that approximates the real $\mathbf{x}_0$ from only $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \alpha_t^{(0)} \mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) + \alpha_t^{(t)} \mathbf{x}_t$$

$$\text{where } \mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$$

Properties of forward and *exact* reverse processes

Property #1:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

$$\text{where } \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$\Rightarrow$ we can sample $\mathbf{x}_t$ from $\mathbf{x}_0$ at any timestep t efficiently in closed form

$\Rightarrow$ $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Property #2: Estimating $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is intractable because of its dependence on $q(\mathbf{x}_0)$. However, conditioning on $\mathbf{x}_0$ we can efficiently work with:

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2\mathbf{I})$$

$$\text{where } \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$$

$$= \alpha_t^{(0)}\mathbf{x}_0 + \alpha_t^{(t)}\mathbf{x}_t$$

$$\sigma_t^2 = \frac{(1-\bar{\alpha}_{t-1})(1-\alpha_t)}{1-\bar{\alpha}_t}$$

Property #3: Combining the two previous properties, we can obtain a different parameterization of $\tilde{\mu}_q$ which has been shown empirically to help in learning p_θ.

Rearranging $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ we have that:

$$\mathbf{x}_0 = \left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right) / \sqrt{\bar{\alpha}_t}$$

Substituting this definition of $\mathbf{x}_0$ into property #2's definition of $\tilde{\mu}_q$ gives:

$$\begin{aligned}\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0) &= \alpha_t^{(0)}\mathbf{x}_0 + \alpha_t^{(t)}\mathbf{x}_t \\ &= \alpha_t^{(0)}\left(\left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right) / \sqrt{\bar{\alpha}_t}\right) + \alpha_t^{(t)}\mathbf{x}_t \\ &= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{(1-\alpha_t)}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}\right)\end{aligned}$$

Parameterizing the *learned* reverse process

Recall: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$

Later we will show that given a training sample $\mathbf{x}_0$, we want

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

to be as close as possible to

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

Intuitively, this makes sense: if the *learned* reverse process is supposed to subtract away the noise, then whenever we're working with a specific $\mathbf{x}_0$ it should subtract it away exactly as *exact* reverse process would have.

Idea #1: Rather than learn $\Sigma_\theta(\mathbf{x}_t, t)$ just use what we know about $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \sim \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2\mathbf{I})$:

$$\Sigma_\theta(\mathbf{x}_t, t) = \sigma_t^2\mathbf{I}$$

Idea #2: Choose μ_θ based on $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$, i.e. we want $\mu_\theta(\mathbf{x}_t, t)$ to be close to $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$. Here are three ways we could parameterize this:

Option C: Learn a network that approximates the $\boldsymbol{\epsilon}$ that gave rise to $\mathbf{x}_t$ from $\mathbf{x}_0$ in the forward process from $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \alpha_t^{(0)}\mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) + \alpha_t^{(t)}\mathbf{x}_t$$

where $\mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) = \left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) / \sqrt{\bar{\alpha}_t}$

where $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$

Parameterizing the *learned* reverse process

Recall: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \mathbf{\Sigma}_\theta(\mathbf{x}_t, t))$

Idea #1: Rather than learn $\Sigma_\theta(\mathbf{x}_t, t)$ just use what we know about $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \sim \mathcal{N}(\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_t^2 \mathbf{I})$:

$$\Sigma_\theta(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$$

Idea #2: Choose μ_θ based on $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$, i.e. we want $\mu_\theta(\mathbf{x}_t, t)$ to be close to $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$. Here are three ways we could parameterize this:

Option A: Learn a network that approximates $\tilde{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$ directly from $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$$

where t is treated as an extra feature in UNet

Option B: Learn a network that approximates the real $\mathbf{x}_0$ from only $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \alpha_t^{(0)} \mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) + \alpha_t^{(t)} \mathbf{x}_t$$

where $\mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$

Option C: Learn a network that approximates the ϵ that gave rise to $\mathbf{x}_t$ from $\mathbf{x}_0$ in the forward process from $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \alpha_t^{(0)} \mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) + \alpha_t^{(t)} \mathbf{x}_t$$

where $\mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) = \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) / \sqrt{\bar{\alpha}_t}$

where $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$

DIFFUSION MODEL TRAINING

Learning the Reverse Process

Recall: given a training sample $\mathbf{x}_0$, we want

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

to be as close as possible to

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

Depending on which of the options for parameterization we pick, we get a different training algorithm.

Option C is the best empirically

Algorithm 1 Training (Option C)

initialize θ
for $e \in \{1, \ldots, E\}$ **do**
    for $x_0 \in \mathcal{D}$ **do**
        $t \sim \text{Uniform}(1, \ldots, T)$
        $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
        $\mathbf{x}_t \leftarrow \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}$
        $\ell_t(\theta) \leftarrow \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2$
        $\theta \leftarrow \theta - \nabla_\theta \ell_t(\theta)$

Option C: Learn a network that approximates the ϵ that gave rise to $\mathbf{x}_t$ from $\mathbf{x}_0$ in the forward process from $\mathbf{x}_t$ and t:

$$\mu_\theta(\mathbf{x}_t, t) = \alpha_t^{(0)} \mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) + \alpha_t^{(t)} \mathbf{x}_t$$

where $\mathbf{x}_\theta^{(0)}(\mathbf{x}_t, t) = \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) / \sqrt{\bar{\alpha}_t}$

where $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) = \text{UNet}_\theta(\mathbf{x}_t, t)$